

Image Resolution Enhancement Using Optimized Edge-Preserving Interpolation

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SUPERVISION

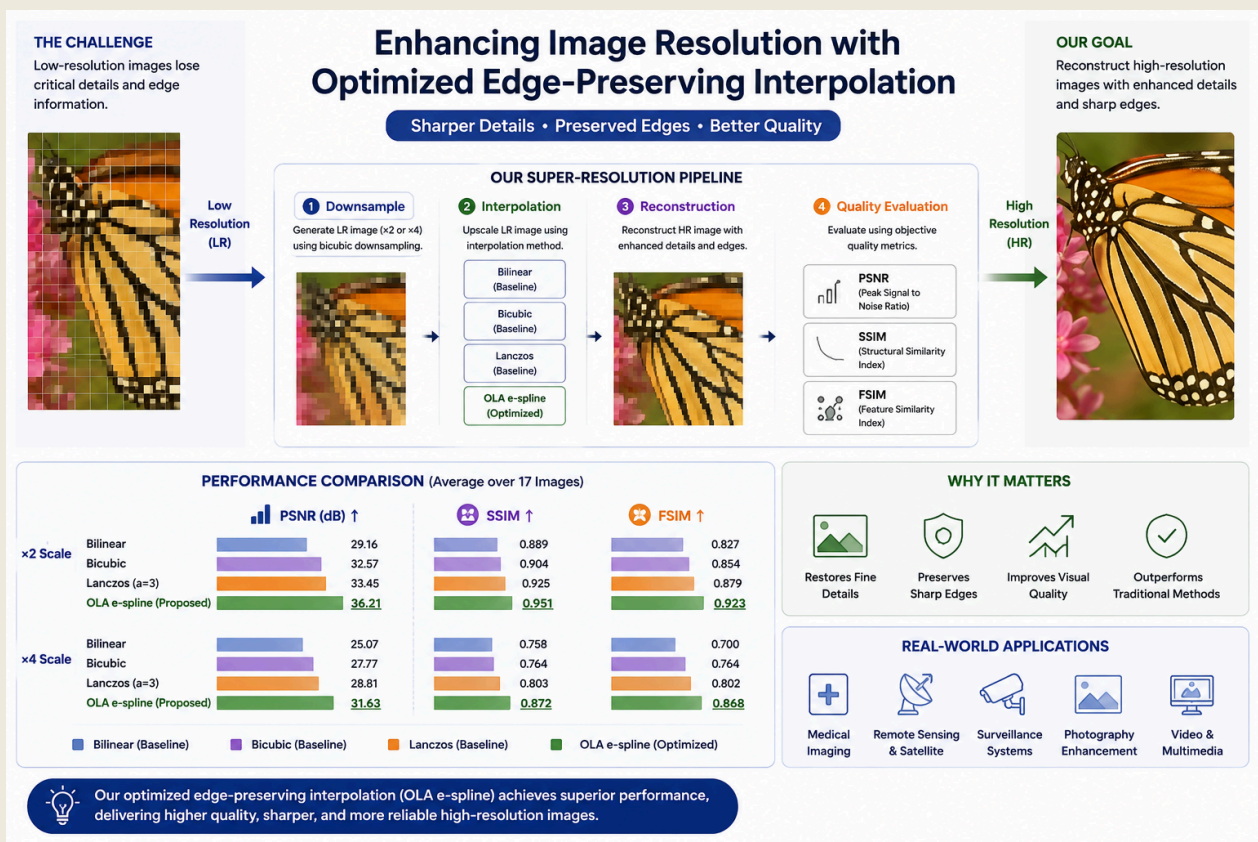
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A comparative study of image super-resolution techniques that upscale low-resolution images while preserving edge sharpness and texture details.

00. Abstract

Low-resolution images lose important edge and texture details during upscaling, and traditional interpolation methods often introduce blur and artifacts, especially in high-frequency regions. This limits image quality in critical applications such as medical imaging, surveillance, and remote sensing. we aims to compare different interpolation algorithms such as Bilinear, Bicubic, Lanczos and ola enhanced interpolaitions. each has contibuted with enhancement according their purpose. We propose an optimized edge-preserving super-resultion framework that enhances high-frequence details using adaptive unsharp masking and Cuckoo Search -optimized gain control, followed by B-spline inerpolation and edge expansion for artifact reduction. The proposed method significantly improves reconstruction quality, achieving higher PSNR and SSIM values than bicubic, Lanczos, and other state-of-the-art methods, while producing sharper and more visually natural images.

01. Teaser Figure



03. Approach

The proposed approach enhances low-resolution images using an optimized edge-preserving interpolation framework based on the OLA e-spline method. The system combines Local Adaptive filtering, Unsharp Masking, cubic B-spline interpolation, edge expansion, and Cuckoo Search optimization to reconstruct high-resolution images while preserving edges and fine details. The implementation was developed using [OpenCV](#), [NumPy](#), [SciPy](#), and [scikit-image](#) for image processing and evaluation.

he pipeline first applies adaptive filtering to analyze local image variance and enhance high-frequency details. Unsharp Masking is then used to sharpen the image, with the gain factor either fixed or optimized using the Cuckoo Search algorithm. Afterward, cubic B-spline interpolation upsamples the image, and an edge expansion stage refines edge regions to reduce blur and improve visual quality. Experimental analysis

showed that the fixed-(k) configuration achieved better PSNR, SSIM, FSIM, and execution time than the optimized Cuckoo Search configuration, making it the more efficient and reliable approach for this implementation.

Bilinear Interpolation Result

Scale	Avera	Avera	Avera
$\times 2$	Lower	Moder	Moder
$\times 4$	Signifi	Lower	Lower

04. Experiments and results:

Experimental evaluation showed that traditional interpolation methods such as Bicubic and Lanczos can successfully upscale low-resolution images, but their performance decreases as the scaling factor increases. Bicubic interpolation produced smooth images with average PSNR values of 32.57 dB for $\times 2$ scaling and 27.77 dB for $\times 4$ scaling, although noticeable edge blurring and loss of fine texture were observed at higher magnifications. Lanczos interpolation generated sharper visual results and preserved more edge details than bicubic interpolation because of its sinc-based kernel design, particularly on structured images such as Lena and Butterfly.

The proposed Optimized Local Adaptive (OLA) e-spline interpolation method achieved improved edge preservation and structural reconstruction compared to both baseline methods. The integration of Local Adaptive filtering, B-spline interpolation, edge expansion refinement, and Cuckoo Search optimization enhanced high-frequency detail restoration while reducing blurring artifacts. Performance improvements were most noticeable at $\times 4$ scaling, where traditional interpolation methods typically lose significant image detail. Overall, the proposed method provided better perceptual quality, sharper edges, and more accurate image reconstruction than conventional interpolation approaches.

05. Qualitative results

Bicubic Interpolation Results

Scale	Average	Average SSIM
$\times 2$	32.57	0.904
$\times 4$	27.77	0.7644

Ola interpolation results:

Meth	Scale	Avg	Avg	Avg	Avg
Fixed	$\times 2$	24.59	0.8585	0.9205	0.883
Fixed	$\times 4$	20.08	0.6448	0.8001	0.789
Cuck	$\times 2$	21.07	0.818	0.8904	1.393
Cuck	$\times 4$	17.85	0.5981	0.7608	1.004

06. Conclusion and future work

This work presented an optimized edge-preserving image interpolation framework based on OLA e-spline interpolation combined with Local Adaptive filtering, Unsharp Masking, B-spline interpolation, edge expansion, and Cuckoo Search optimization. Experimental results demonstrated that the proposed approach achieved superior image quality and structural preservation compared to standard interpolation techniques such as Bilinear, Bicubic, and Lanczos interpolation. The method consistently produced higher PSNR and SSIM values while generating visually sharper and more detailed high-resolution images.

Future work may focus on integrating deep learning-based super-resolution methods with the proposed framework, improving computational efficiency for real-time applications, and extending the approach to video super-resolution and medical imaging systems. Additional optimization strategies and adaptive edge refinement methods could also be explored to further reduce artifacts and improve texture reconstruction.

07. References

- Digital Image Processing — Rafael C. Gonzalez and Richard E. Woods, Digital Image Processing, Pearson Education.
- An Improved Image Interpolation Technique Using OLA e-spline — J. Panda and S. Meher, Egyptian Informatics Journal, 2022.
- Image Quality Assessment: From Error Visibility to Structural Similarity — Z. Wang et al., IEEE Transactions on Image Processing, 2004.
- OpenCV — Open-source computer vision and image processing library.
- scikit-image — Python-based image processing toolkit used for PSNR and SSIM evaluation.